

## SOFTWARE ENGINEERING – LAB 01

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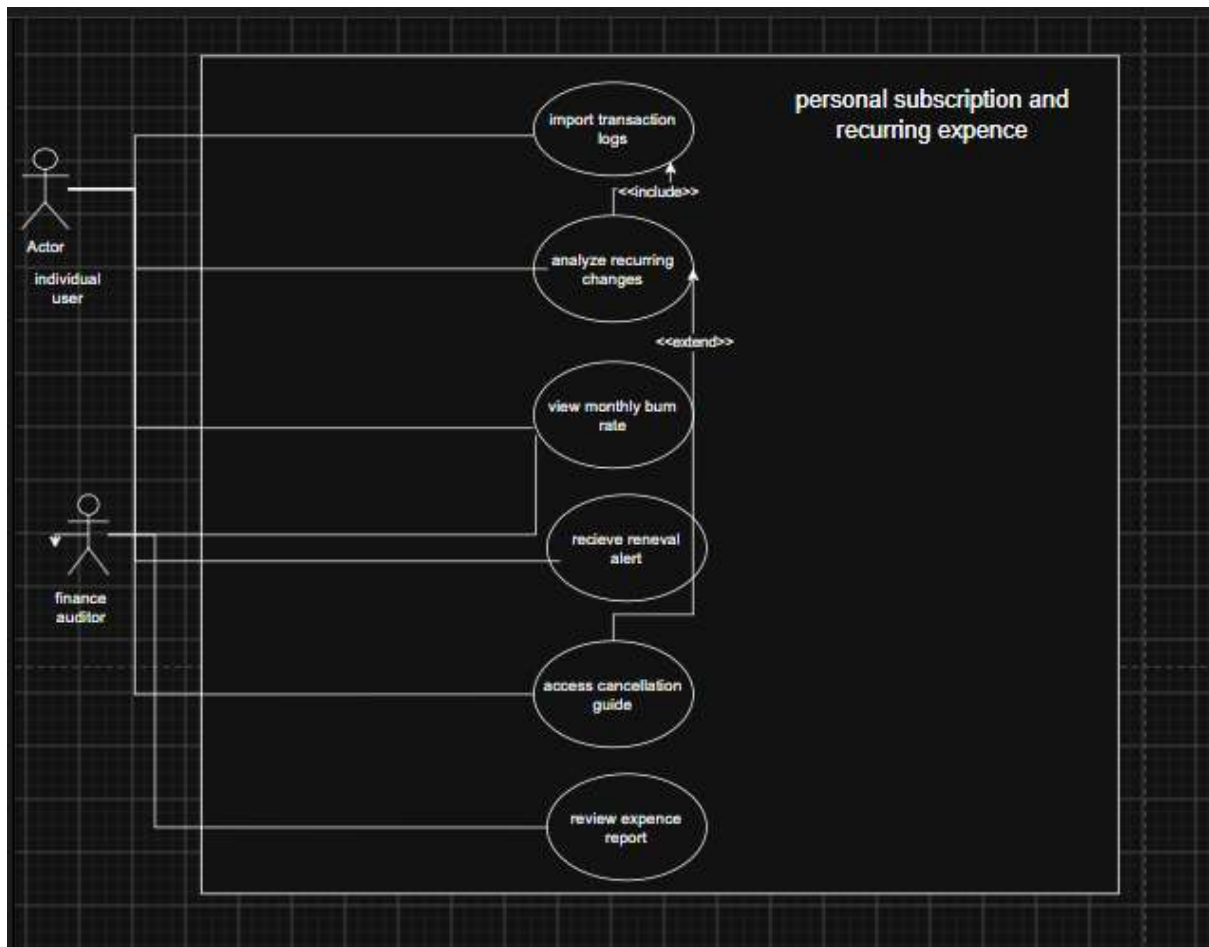
SECTION : K

| ID       | TYPE       | DESCRIPTION                                                                                                                          | PRIORITY | Acceptance criteria                                                                                                                                                                                    | Rationale                                                                              | comments                                                                   |
|----------|------------|--------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| FR - 001 | FUNCTIONAL | The system shall analyze transaction logs to identify recurring billing patterns and project total monthly subscription expenditure. | high     | <b>Pass:</b> Recurring service identified and renewal date logged.<br><b>Fail:</b> One-time payment misclassified as a subscription                                                                    | Core functionality for identifying recurring expenses and estimating monthly spending. | Recurring transactions are detected and subscription details are recorded. |
| FR- 002  | FUNCTIONAL | Core functionality for identifying recurring expenses and estimating monthly spending.                                               | high     | <b>Pass:</b> A valid CSV or OFX file is successfully imported and transaction records are available for analysis. <b>Fail:</b> An invalid or unsupported file format is rejected with an error message | Transaction data is required for the system to identify recurring expenses.            | Supports importing transaction data in OFX format                          |

|         |                |                                                                                                                                               |        |                                                                                                                                                                                                                        |                                                                         |                                                                   |
|---------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------|
| FR-003  | FUNCTIONAL     | The system shall alert the user before the renewal date of an identified subscription.                                                        | high   | <b>Pass:</b> The system sends an alert containing the subscription name and upcoming renewal date before the renewal date.<br><b>Fail:</b> No alert is sent before the renewal date.                                   | Helps users avoid unwanted subscription renewals and recurring charges. | Alerts users of upcoming subscription renewals.                   |
| FR- 004 | FUNCTIONAL     | The system shall calculate and display the user's total monthly burn rate based on identified recurring subscriptions and recurring expenses. | high   | <b>Pass:</b> The displayed monthly burn rate equals the sum of all identified monthly recurring expenses.<br><b>Fail:</b> The calculated total does not match the identified recurring expenses.                       | Helps users understand how much they spend regularly each month.        | Provides a clear view of monthly expenses.                        |
| FR-005  | FUNCTIONAL     | The system shall provide a one-click cancellation guide for each identified subscription.                                                     | medium | <b>Pass:</b> When the user selects the cancellation option, the system displays the relevant steps or link to cancel the subscription. <b>Fail:</b> No cancellation guide is displayed or an incorrect guide is shown. | Helps users easily manage and cancel unwanted subscriptions.            | Provides easy-to-follow instructions for subscription management. |
| NFR-001 | NON FUNCTIONAL | Financial transaction imports (CSV/OFX)                                                                                                       | high   | <b>Pass:</b> Benchmarking tests confirm target latency and security standards under                                                                                                                                    | Protects sensitive                                                      | Ensures security of financial data.                               |

|         |                |                                                                                                                                 |        |                                                                                                                                                                                                  |                                                                             |                                       |
|---------|----------------|---------------------------------------------------------------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------|
|         |                | must parse securely without storing raw banking credentials in cleartext.                                                       |        | simulated peak load.<br><b>Fail:</b> Raw banking credentials are stored in cleartext or the system fails to meet the defined performance/security standards.                                     | financial information and ensures secure, efficient transaction processing. |                                       |
| NFR-002 | NON FUNCTIONAL | The system shall display the user's subscription and recurring expense information in a clear and easy-to-understand interface. | medium | <b>Pass:</b> Users can view subscription name, recurring amount, renewal date, and monthly burn rate without assistance.<br><b>Fail:</b> Required information is unclear or difficult to locate. | Ensure users can easily understand and manage their recurring expenses.     | Focuses on clear, and user interface. |

Use case diagram :



Use case flow:

#### Use-Case Flow Specification

#### Use Case: Analyze Recurring Transactions

**Use Case ID:** UC-01

**Primary Actor:** Individual User

**Supporting Actor:** Finance Auditor

#### Preconditions

1. The user is logged into the Personal Subscription & Recurring Expense Auditor.
2. The user has imported a valid transaction file such as CSV or OFX.
3. The transaction data contains valid payment details required for analysis.

#### Postconditions

1. Recurring subscriptions are identified and displayed to the user.
2. Renewal dates are recorded where they can be determined.
3. The system calculates and displays the projected monthly subscription expenditure.

4. The user can review the identified recurring expenses.

#### **Main Success Scenario**

1. The Individual User selects **"Analyze Recurring Transactions."**
2. The system retrieves the imported transaction records.
3. The system validates the transaction data.
4. The system groups transactions based on recurring billing patterns such as merchant, amount, and billing frequency.
5. The system identifies transactions that match recurring subscription patterns.
6. The system estimates the next renewal date for each identified subscription.
7. The system calculates the projected total monthly subscription expenditure.
8. The system displays the identified subscriptions, renewal dates, and monthly expenditure to the user.
9. The user reviews the recurring expense report.
10. The use case ends successfully.

#### **Alternate Flow – One-Time Payment Detected**

**5a.** If a transaction does not have sufficient evidence of a recurring billing pattern, the system does not classify it as a subscription.

**5a1.** The system marks the transaction as a one-time payment or excludes it from recurring expenses.

**5a2.** The system continues analyzing the remaining transactions.

**5a3.** The system displays the final recurring subscription list and monthly expenditure without the misclassified payment.